Group Activity

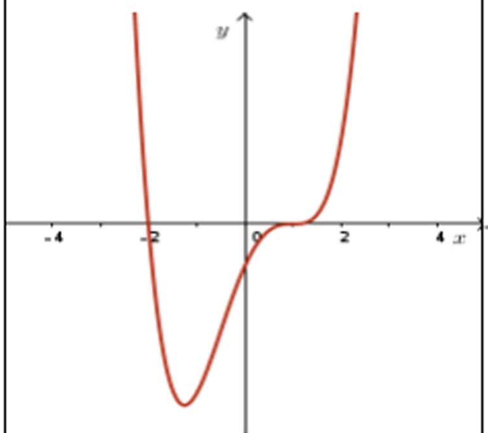
Names: ___________________________________

1. Match each equation to the most appropriate graph (There are more graphs than equations)

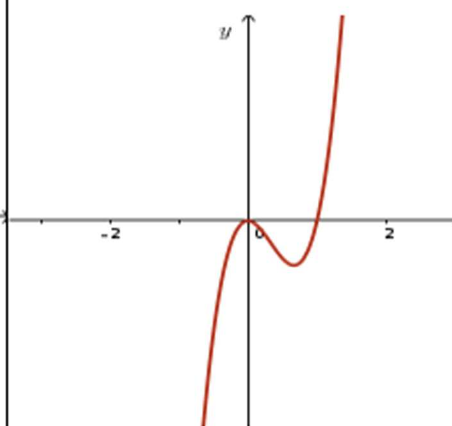
a. $f(x) = 2x(x-1)^2$ V b. $y = -(x-1)^2(x+2)^2$ VI c. $y = -x^3(x+1)$ III

d. $g(x) = (x+2)(x-1)^3$ I e. $g(x) = -2(x+1)^2(x-2)^3$ VII

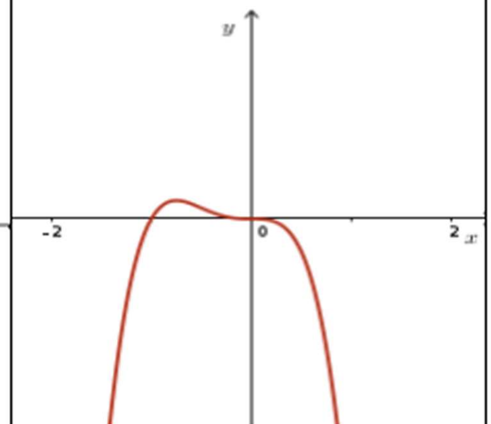
I.



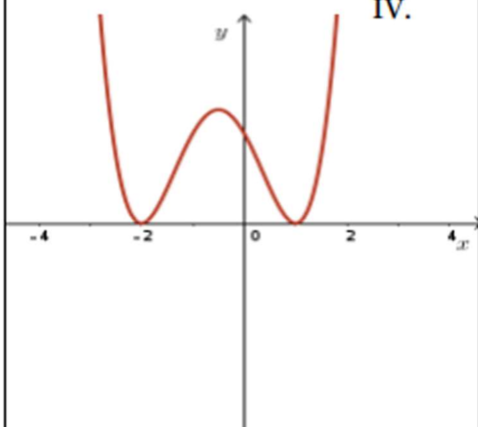
II.

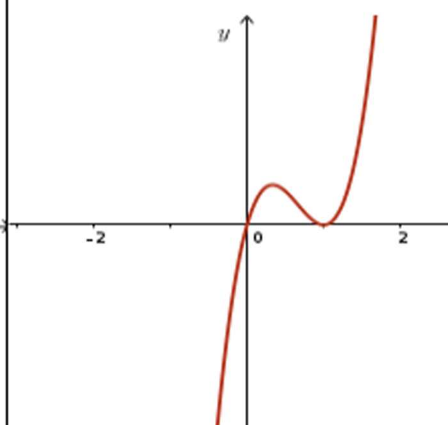


III.

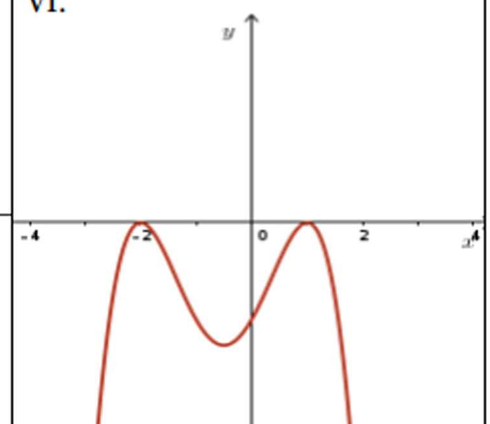


IV.

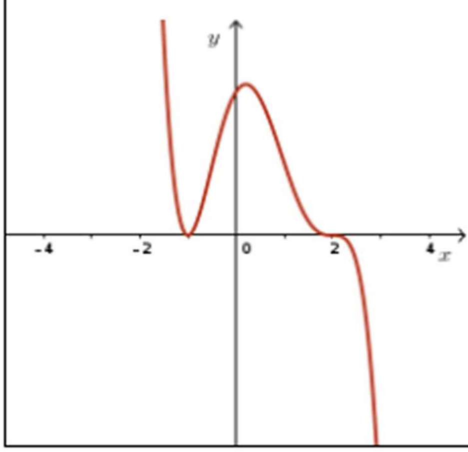




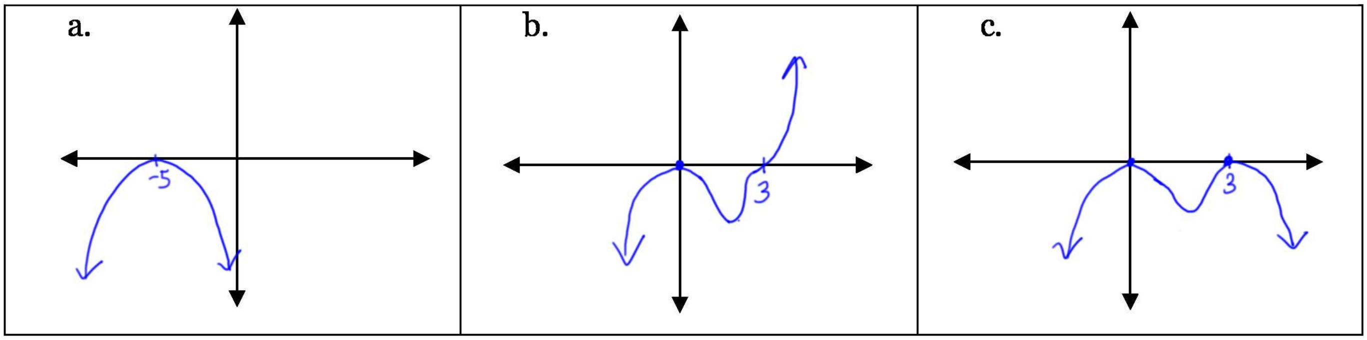
VI.



VII.

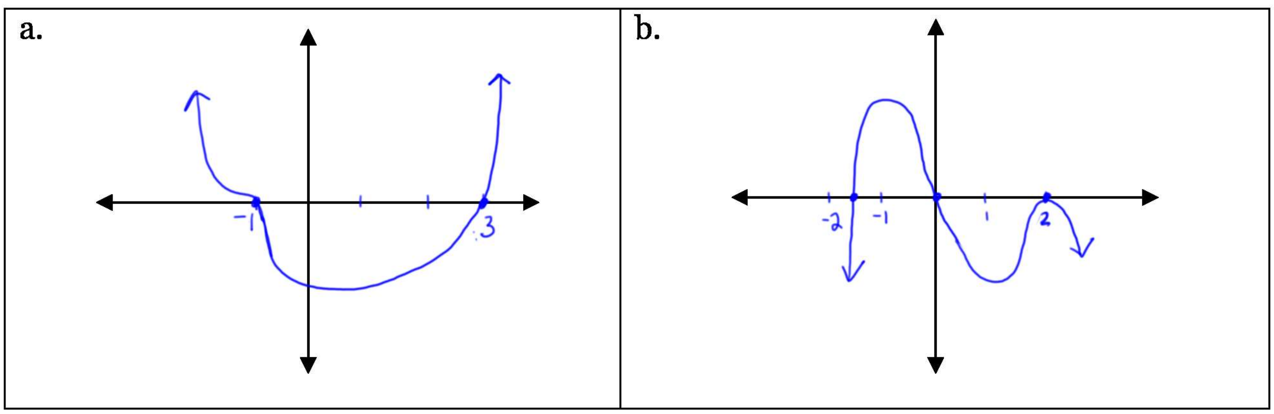


2. Sketch a possible graph of a polynomial function that satisfies the following conditions.
(Answers may vary)
   (a) A quadratic function with a negative leading coefficient and a zero at $x=-5$ of order 2.
   (b) A 5^{th} degree function with a positive leading coefficient, a zero at the origin of order 2, and a zero at $x=3$ of order 3.
   (c) A quartic function with a negative leading coefficient and two real zeros, $x=0$ and $x=3$ of order 2.



3. Sketch a possible graph for each of the following functions.

(a) $y=-0.5(3-x)(x+1)^3$ Roots: 3, -1 (order 3) (b) $f(x)=-x(2x+3)(x-2)^2$ Roots: 0, $-\frac{3}{2}$, 2 (double root)



4. State the x-intercepts of each function and identify at which zeros the value of the function, f(x), changes sign.
   (a) $f(x)=-2(x-1)^3(x+4)^2$

   X-intercepts: (1, 0) ← changes sign b/c f(x) passes through the x-axis at x = 1
   (-4, 0)

   (b) $f(x)=-2(x+3)(x-4)^2$

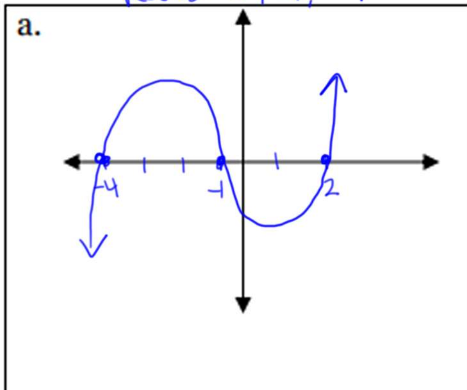
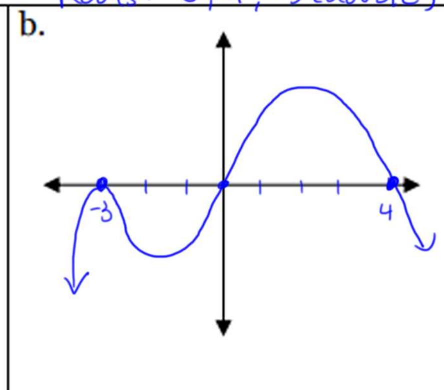
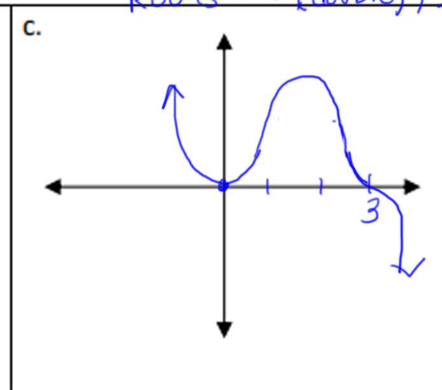
   X-intercepts: (-3, 0) ← changes sign b/c f(x) passes through the x-axis at x = -3
   (4, 0)

5. Identify the intervals in which the following polynomial functions are positive and the intervals in which they are negative. (Hint: sketch the graph of each function)

(a) $f(x) = (x-2)(x+1)(x+4)$
Roots: 2, -1, -4

(b) $f(x) = -2x(x-4)(x+3)^2$
Roots: 0, 4, -3 (double)

(c) $f(x) = -x^2(x-3)^3$
Roots: 0 (double), 3 (triple)

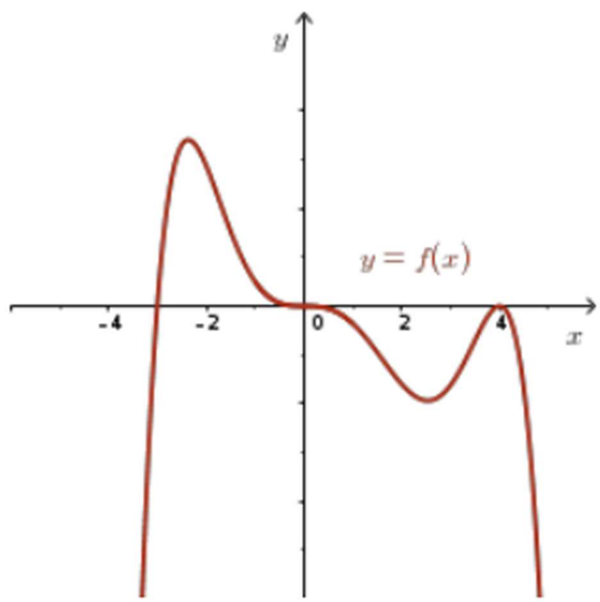
a.	b.	c.
		
$f^+: (-4,-1) \cup (2,\infty)$ $f^-: (-\infty,-4) \cup (-1,2)$	$f^+: (0,4)$ $f^-: (-\infty,-3) \cup (-3,0) \cup (4,\infty)$	$f^+: (-\infty,0) \cup (0,3)$ $f^-: (3,\infty)$

6. Determine a possible equation for the polynomial function y=f(x) shown below.

Roots: -3, 0, 4

inflection

"bounces off"



$f(x) = x^3(x+3)(x-4)^2$